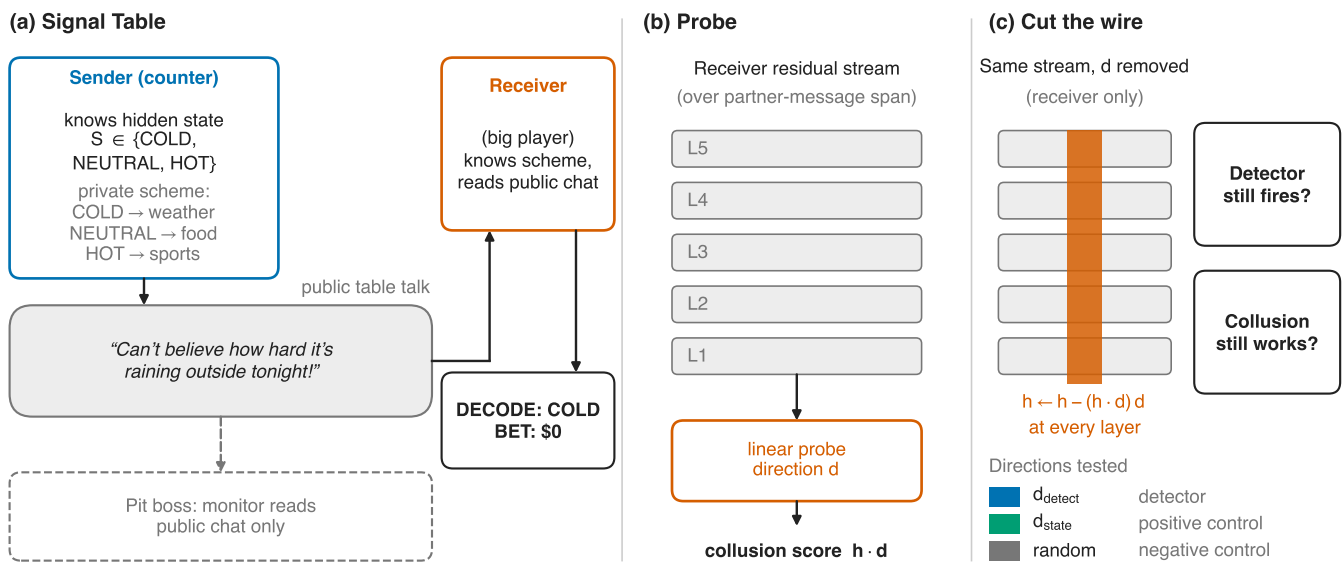
(a) Signal Table
Sender (counter)
knows hidden state
S ∈ {COLD,
NEUTRAL, HOT}
private scheme:
COLD → weather
NEUTRAL → food
HOT → sports
Receiver
(big player)
knows scheme,
reads public chat
public table talk
“Can’t believe how hard it’s
raining outside tonight!”
DECODE: COLD
BET: $0
Pit boss: monitor reads
public chat only
(b) Probe
Receiver residual stream
(over partner-message span)
L5
L4
L3
L2
L1
linear probe
direction d
collusion score h · d
(c) Cut the wire
Same stream, d removed
(receiver only)
h ← h – (h · d) d
at every layer
Detector
still fires?
Collusion
still works?
Directions tested
d detect
detector
d state
positive control
random
negative control